

Problem 3

Proposition
from Lecture 10
(in particular (c))

Fact (1) $\alpha \in k$: finite field of char. p
 $|k| = p^d$

$$\text{Fr}^m(\alpha) = \alpha \quad (\Leftrightarrow) \quad \text{Fr}^{\gcd(m, d)}(\alpha) = \alpha$$

Fact (2): $|k| = p^d$, $f(x) \in \mathbb{F}_p[x]$ irred.
of deg r
(Problem 2)

TFAB: (1) $r \mid d$

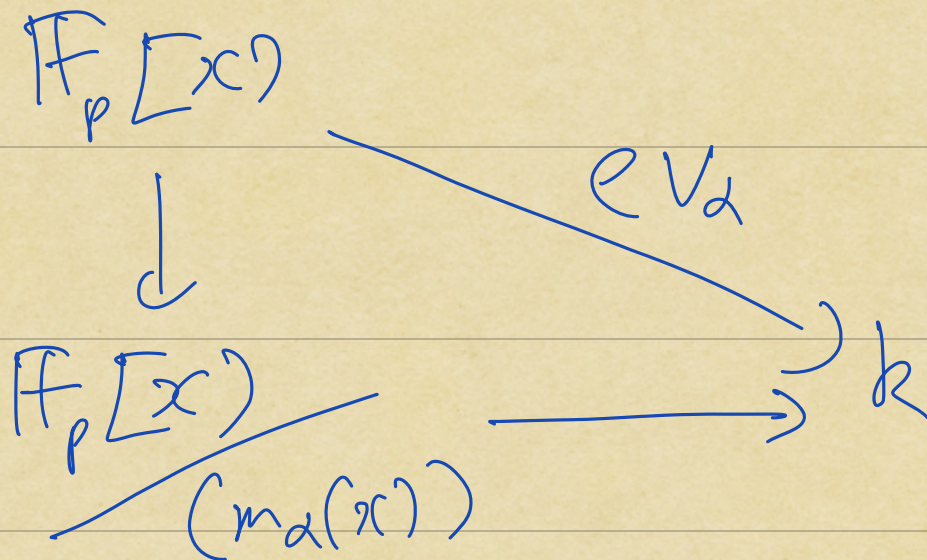
(2) $f(x)$ has r distinct
zeros in k

(3) $f(x)$ has c zero
in k .

$\alpha \in k$: finite of char p

Defⁿ $m_\alpha(x)$ is the irreducible polynomial with leading coeff 1 s.t.

$$(m_\alpha(x)) = \ker \text{ev}_\alpha$$



Fact (3): TFAE for $f(x) \in \mathbb{F}_p[x]$

- (1) $f(\alpha) = 0 \in k$
- (2) $f(x) \in \ker \text{ev}_\alpha$
- (3) $m_\alpha(x) \mid f(x)$

Linear algebra

k : field

Defⁿ

A k -vector space

or a vector space over k

is a tuple $(V, +, 0, \cdot)$

where:

(a) $(V, +, 0)$ is an additive abelian group

(Scalar multiplication) $\rightarrow (b) \cdot : k \times V \longrightarrow V$
 $(a, v) \mapsto a \cdot v$

S.t. (1) (Distributivity)
 $a \in k, v_1, v_2 \in V$

$$a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$$

(2) (Associativity)
 $a, b \in k, v \in V$

$$a \cdot (b \cdot v) = (ab) \cdot v$$

(3) (Identity for scalar multiplication)

$$\forall v \in V$$

$$1 \cdot v = v$$

Examples

① $\{0\}$: k -vector space

with

$$a \cdot 0 = 0, \forall a \in k.$$

② k :

$$k \times k \longrightarrow k$$

$$(a, b) \longmapsto a \cdot b$$

③ V, W : k -vector spaces

$V \times W$: direct product
is also a k -vector space with

$$\text{R.H.S. } a \in k^*$$

$$a^{-1} \cdot (a \cdot v)$$

$$= (a^{-1}a) \cdot v$$

$$= 1 \cdot v$$

$$= v$$

(0-ordinate wise addition &
scalar multiplication

$$(4) \quad \underbrace{k \times \dots \times k}_{n\text{-times}} \xrightarrow{\text{def}} k^n$$

$$k^n = \{(a_1, a_2, \dots, a_n) : a_i \in k\}$$

$$(5) \quad k[x] : k\text{-vector space}$$

$$k \times k[x] \longrightarrow k[x]$$

$$(a, f(x)) \mapsto a \cdot f(x)$$

$$(6) \quad k \underset{\substack{\uparrow \\ \text{Sub-ring}}}{\leq} R \leftarrow \text{commutative ring}$$

then R is a k -vector space

$$\begin{aligned} k \times R &\longrightarrow R \\ (a, r) &\mapsto a \cdot r \end{aligned}$$

(7) In the situation of (6)
 $I \leq R$: ideal

Then

I is a k -subspace

of R
(8) In particular, $k \leq L$: field ext.
then L is a k -vector space.

Defⁿ V : k -vector space

$W \leq V$: subgroup

W is a k -subspace (or just subspace)

if $\forall a \in k, w \in W,$

$a \cdot w \in W$

(i.e. closed
under $\cdot$)

Examples:

(1) $\{0\} \leq V$
subspace

(2) $V \leq V$
subspace

Rmk: $k \subseteq L$: field ext.

Then any L -vector space
also has the structure of
a k -vector space.

Defⁿ: V : k -vector space

$X \subseteq V$: subset

The k -subspace generated by X

$$\langle X \rangle_k \leq V$$

is the smallest subspace of V
containing X

i.e. If $W \leq V$ is any subspace
containing X , then

$$\langle X \rangle \leq W.$$

Rmk: (1) $W_i \leq V$ are k -subspaces
 $i \in I$ (I is an indexing set)

Then

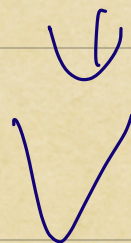
$$\bigcap_{i \in I} W_i \leq V$$

is also a k -subspace

$$(2) \quad \underline{I}_X = \left\{ \begin{array}{l} \text{Subspaces } W \leq V \\ \text{s.t. } X \leq W \end{array} \right\}$$

$(X \leq V)$

(top-down)



$$\bigcap_{W \in I_X} W = \underline{\langle X \rangle}.$$

Defn $v_1, v_2, \dots, v_n \in V$

A k -linear combination

of $v_1, \dots, v_n$ is an element of the form $v = a_1 v_1 + \dots + a_n v_n \in V$ with $a_1, a_2, \dots, a_n \in k$

(3) (Bottom-up)

$$\langle X \rangle = \left\{ \sum_{i=1}^n a_i v_i : \begin{matrix} a_i \in k \\ v_i \in X \end{matrix} \right\}$$

= { k -linear combinations
of elements of X }

Fact V is k -v.s.

(a) $v \in V$

$$0 \cdot v = 0_V \in V$$

(b) $v \in V$

$$(-1) \cdot v = -v \in V$$

Pf: Enough to show:

(a) RHS is a λ -subspace
of V

(b) RHS is contained
in every subspace containing X .